

Original Scientific Paper

Myocardial sarcoplasmic reticulum Ca^{2+} ATPase function is increased by aerobic interval training

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Received 18 May 2007 Accepted 9 July 2007

Objective Reduced activity of the sarcoplasmic reticulum Ca^{2+} ATPase-2a (SERCA-2a) contributes to myocardial dysfunction. Exercise training improves myocardial Ca^{2+} -handling, but SERCA-2a function is uncertain. We assessed SERCA-2a activity after exercise training.

Methods SERCA-2a function was assessed by sarcoplasmic reticulum Ca^{2+} uptake in cardiomyocytes with other Ca^{2+} uptake mechanisms blocked, in mice after aerobic interval training versus sedentary controls.

Results We established protocols to assess SERCA-2a function, and show that aerobic interval training increases the maximal rate of Ca^{2+} uptake by 30%. This is at least partly explained by reduced phospholamban-to-SERCA-2a ratio.

Conclusion Aerobic interval training improves myocardial SERCA-2a performance, explaining at least partly why myocardial Ca^{2+} -handling improves after exercise training. *Eur J Cardiovasc Prev Rehabil* 15:145–148 © 2008 The European Society of Cardiology

European Journal of Cardiovascular Prevention and Rehabilitation 2008, 15:145–148

Keywords: calcium, exercise, myocardium, sarcoplasmic reticulum Ca^{2+} ATPase

Introduction

The myocardium adapts to regular exercise training by improving contractile performance. At the cellular level, contraction is regulated by the free intracellular Ca^{2+} concentration ($[\text{Ca}^{2+}]_i$), and improved handling of Ca^{2+} after exercise training leads to improved contractile performance of the cardiomyocyte [1–5]. Notably, faster rates of Ca^{2+} cycling have been reported repeatedly, especially during diastole [2–5]. This suggests that the performance of one or more Ca^{2+} sequestering mechanisms in the cardiomyocyte adapt to exercise training. Of particular interest is the sarcoplasmic reticulum (SR) Ca^{2+} ATPase-2a (SERCA-2a) that resequesters the majority of cytosolic Ca^{2+} back to the SR during diastole on a beat-to-beat basis [6]. Although protein expression

studies have reported that SERCA-2a in the myocardium increases after exercise training without increased phospholamban (PLB), the innate negative regulator of SERCA-2a [4,5,7], the activity of SERCA-2a and hence rate of SR Ca^{2+} uptake remains unknown after exercise training.

Here, we report an experimental approach to study SERCA-2a function by Fura-2 based fluorescence microscopy, and use this approach to show that regular aerobic interval training improves SERCA-2a function in the myocardium, at least partly owing to reduced PLB-to-SERCA-2a ratio.

Methods

Animals and exercise training

Twenty female C57BL/6J mice (Møllegaards Breeding Center, Lille Skensved, Denmark), 19–21 g and 8 weeks

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old at inclusion were randomized to sedentary control or regular aerobic interval training on an inclined (25°) treadmill for 6 weeks, 5 days/week, 1.5 h/day, at 85–90% of maximal oxygen uptake ($\dot{V}O_{2\max}$). Protocols for animal handling and exercise training have been reported previously [8]. Moreover, 10 mice were used to establish the protocols for measuring SERCA-2a function. Experimental protocols were approved by the Animal Research Ethics Committee.

SERCA-2a function

Left ventricular tissue samples were excised and permeabilized by saponin (50 µg/ml, 30 min, 4°C), before being transferred to a solution containing (in mmol/l) 0.05 ethylene glycol tetraacetic acid, 5 ATP, 10 CrP, 25 HEPES, 100 KCl, and 5.5 MgCl; pH 7.0. Fura-2 (10 µmol/l; Molecular Probes, Eugene, Oregon, USA) to probe Ca^{2+} , oxalate (10 µmol/l; Sigma, St Louis, Missouri, USA) to stabilize intra-SR [Ca^{2+}], and ruthenium red (3 µmol/l; Sigma) to inhibit SR Ca^{2+} efflux and mitochondrial Ca^{2+} buffering, were added. The experiment was initiated by stirring tissue samples in a 150-µl cuvette while continuously monitoring extra-SR Ca^{2+} on an inverted epifluorescence microscope (Diaphot-TMD, Nikon, Tokyo, Japan) by counting 510-nm emission after exciting with alternating 340 and 380-nm wavelengths (Photon Technology International, Lawrenceville, New Jersey). [Ca^{2+}] was rapidly increased by 100 µmol/l, and the subsequent decline was interpreted as SERCA-2a transport (SR Ca^{2+} uptake), as other Ca^{2+} pathways were 'removed.' Thapsigargin (10 µmol/l; Sigma) was added at the end of the protocol to assay any SR Ca^{2+} leakage. The ratio signal was converted to total [Ca^{2+}] by measuring $R_{\min}$ (≤ 10 nmol/l Ca^{2+}) and $R_{\max}$ (375 nmol/l Ca^{2+}) in each tissue sample, and assuming a K_d of 250 nmol/l for the affinity of Fura-2 for Ca^{2+} , in agreement with Gryniewicz *et al.* [9] and as obtained in our laboratory. Subsequently, the rate of SR Ca^{2+} uptake was plotted against the corresponding free [Ca^{2+}], and a logistic curve was finally fitted to estimate maximal Ca^{2+} uptake rate ($V_{\max}$) and the free [Ca^{2+}] that evoked half Ca^{2+} uptake rate (K_m); both corrected for protein content as measured by mass spectrometry in each sample.

SERCA-2a and phospholamban expression

Frozen left ventricular samples were homogenized in lysis buffer (30 mmol/l Tris-HCl, 150 mmol/l KCl, 300 mmol/l sucrose, and 10 mmol/l NaF; pH 7.5) and centrifuged for 20 min at 8000g and 30 min at 100 000g at 4°C, whereupon the pellet was considered as the SR/membrane-enriched fraction. Thirty-microgram extracts from trained and sedentary hearts were loaded on 15% sodium dodecyl sulphate-polyacrylamide gel electrophoresis. Proteins were blotted on nitrocellulose membranes (Bio-Rad Laboratories, Hercules, California, USA) that were blocked with Tris-buffered saline-T/milk (50-mmol/l

Tris-HCl, 150 mmol/l NaCl, 0.1% Tween-20, 5% w/v nonfat dried milk, pH 7.4) for 1 h at room temperature, and thereafter incubated for 1 h in room temperature with the specific antibodies for SERCA and PLB (Badrilla, Leeds, UK). Horseradish peroxidase-conjugate goat antirabbit and antimouse were used as secondary antibodies (Amersham, Buckinghamshire, UK). Finally, membranes were developed with the enhanced chemiluminescence detection system (Amersham).

Statistics

The Mann-Whitney *U*-test detected differences, with significance level $P < 0.05$.

Results

Aerobic interval training over 6 weeks increased $\dot{V}O_{2\max}$ by ~40%, from 33 ± 2 to 46 ± 2 ml/kg^{0.75}/min, whereas no change occurred in sedentary mice.

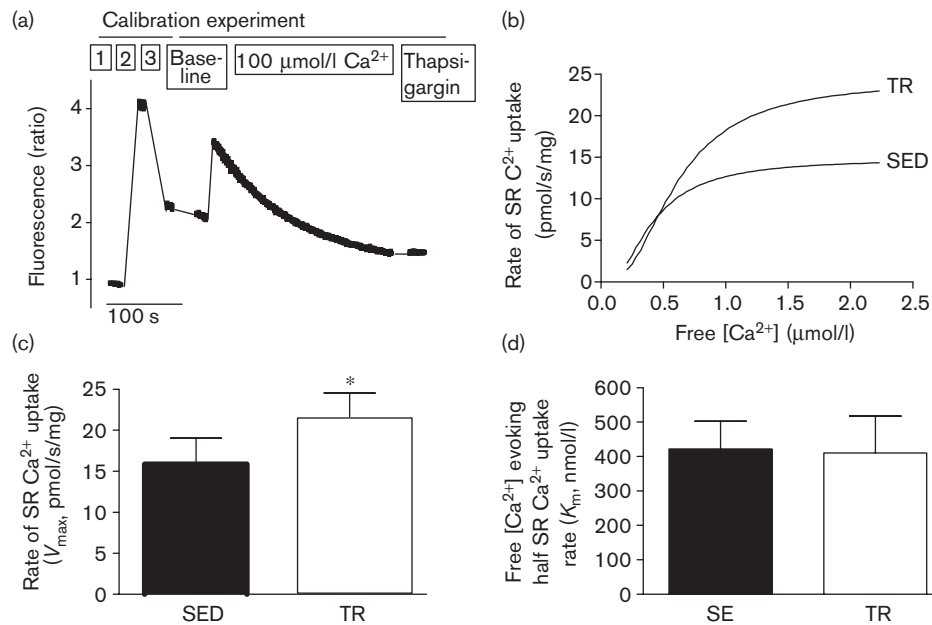
To study diastolic function in more detail, we developed a Fura-2 fluorescence-based microscopy technique to assess SR Ca^{2+} uptake in the myocardium; hence the function of the SERCA-2a. Figure 1 shows a typical trace of the SR Ca^{2+} uptake measurements, the logistically fitted curve for estimating Ca^{2+} uptake rate against free [Ca^{2+}], and the training-induced effects on $V_{\max}$ and K_m . Here, we show that the maximal rate of SR Ca^{2+} uptake ($V_{\max}$) by the SERCA-2a was ~30% higher in the myocardium after aerobic interval training, as opposed to myocardium from sedentary control mice. In contrast, we did not detect any training-induced changes to the free [Ca^{2+}] that evoked half-maximum SR Ca^{2+} uptake rate (K_m).

Higher maximal activity of SERCA-2a was at least partly explained by a 10% ($P < 0.01$) increase in the protein level of SERCA-2a and unchanged abundance of total PLB in the SR, as determined by immunoblots of SR fractions of the myocardium. Hence, the PLB-to-SERCA-2a ratio decreased with exercise training (Fig. 2). Finally, we plotted SERCA-2a activity with expression to illustrate the relationship between protein and function, and how this adapts to exercise training; see Fig. 2c.

Discussion

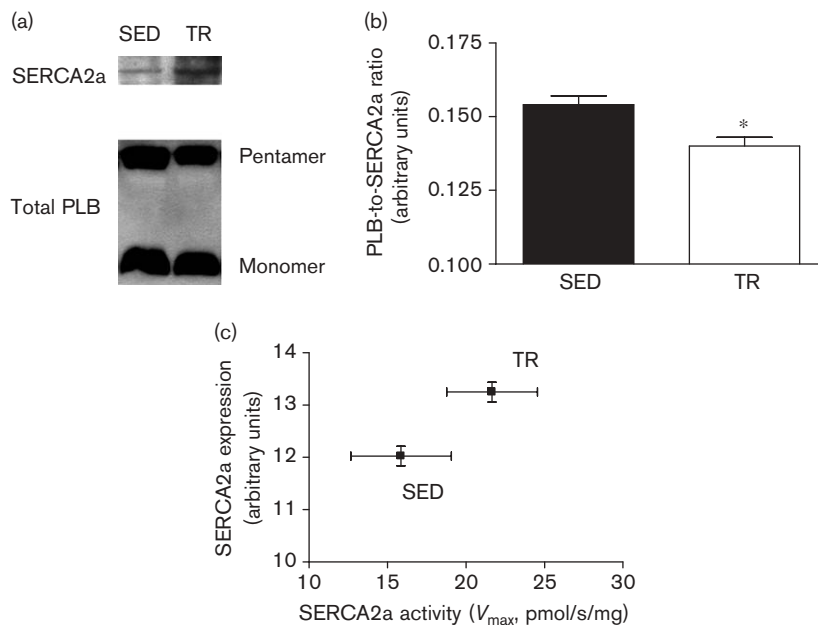
This study demonstrates that myocardial SERCA-2a function is enhanced by a program of aerobic interval training at high exercise intensity, as the maximal rate of Ca^{2+} uptake into the SR (SERCA-2a $V_{\max}$) in the myocardium increased. Hence, improved SERCA-2a function is a likely explanation for the faster rates of the Ca^{2+} transient decay, that is, improved diastolic Ca^{2+} handling in trained cardiomyocytes that leads to faster relaxation rates of the cell [2–5,10], and to the lower diastolic [Ca^{2+}]; reported previously [7]. It is also likely that improved SERCA-2a function affects inotropy, owing to prolonged ventricular filling time and higher

Fig. 1



(a) Representative trace of a sarcoplasmic reticulum (SR) Ca^{2+} ATPase (SERCA-2a) Ca^{2+} uptake measurement; calibration solutions 1, 2, and 3 refer to $R_{\min}$ ($\leq 10 \text{ nmol/l } \text{Ca}^{2+}$), $R_{\max}$ ($375 \text{ nmol/l } \text{Ca}^{2+}$), and a 50/50 solution of both, respectively. (b) The fitted logistic curves for estimation of Ca^{2+} uptake rate from trained (TR) and sedentary (SED) mice. (c) Rate of maximal Ca^{2+} uptake ($V_{\max}$) in the SR. (d) Free intracellular $[\text{Ca}^{2+}]$ ($[\text{Ca}^{2+}]_i$) that evokes 50% of the Ca^{2+} uptake rate (K_m). Panels (c) and (d) show means $\pm$ SD, * $P < 0.05$.

Fig. 2



(a) Representative immunoblots of protein expression levels of sarcoplasmic reticulum Ca^{2+} ATPase (SERCA-2a) and total pentameric and monomeric phospholamban (PLB) from trained (TR) and sedentary (SED) mice. (b) PLB-to-SERCA-2a protein expression ratio. (c) Correlation plot of SERCA-2a activity and expression levels. Panels (b) and (c) show means $\pm$ SD, * $P < 0.01$.

loading of Ca^{2+} in the SR. The mechanism for improved SERCA-2a function appears at least partly to be due to increased abundance of protein levels of SERCA-2a in the heart, as also reported previously [7]. As SERCA-2a expression levels increase, but PLB expression does not change, it means that the PLB-to-SERCA-2a ratio decreases by exercise training (i.e. SERCA-2a is less inhibited). The reason for the y -intercept being different than 0 at $x = 0$ is probably a reflection of a nonlinearity between protein levels and the immunoblot signals, and the finding that exercise training induces constitutive phosphorylation of PLB at the threonine 17 residue [7], which also would contribute to improved SERCA-2a function. As the contribution of SERCA-2a in the removal of cytosolic Ca^{2+} during diastole is species-specific; $\sim 92\%$ in mouse versus $\sim 70\%$ in man [6], it is conceivable that quantitative, rather than qualitative, differences exist between different species as to the magnitude of the training-induced adaptation.

The experimental approach to assess SERCA-2a function was adapted from spectrophotometer protocols [11] to the epifluorescence imaging system described in this report. The protocol traps Ca^{2+} inside the SR once it is taken up by the SERCA-2a, owing to the oxalate and ruthenium red maneuvers to retain Ca^{2+} in the SR and block the ryanodine receptor (RyR-2). Ruthenium red has also been shown to inhibit SERCA-2a and quench Fura-2 [12], but these effects are minimal at low concentrations ($3 \mu\text{mol/l}$), and appear to affect both excitation wavelengths equally. Moreover, ruthenium red blocks mitochondrial Ca^{2+} uptake, whereas the permeabilization of the cell membrane removes the contributions of the $\text{Na}^+/\text{Ca}^{2+}$ exchanger and the plasma membrane Ca^{2+} ATPase. Hence, the SERCA-2a is the only remaining route that Ca^{2+} might take in this preparation. The reduced signal from the Fura-2 Ca^{2+} is therefore interpreted as SR Ca^{2+} uptake by SERCA-2a. As an extra control at the end of each experiment, we added thapsigargin to block SERCA-2a and to assay any RyR-2 leak. The flat Fura-2 Ca^{2+} signal observed suggests that the SR remains intact during the experiments, traps Ca^{2+} inside, and that SERCA-2a is the only contributor to the reduced free $[\text{Ca}^{2+}]$ in the preparation.

Conclusion

This study demonstrated that the maximal rate of SR Ca^{2+} uptake, hence the maximal activity of SERCA-2a, in the myocardium is increased after a program of aerobic

interval training. At least partly, the improved function is due to reduced PLB-to-SERCA-2a ratio in exercised myocardium. This suggests that improved SERCA-2a function explains at least partly the previous reports of improved diastolic Ca^{2+} handling and lower diastolic $[\text{Ca}^{2+}]$ after exercise training in the heart.

Acknowledgements

This study was supported by grants from the Norwegian University of Science and Technology (O.J.K., U.W.), the St Olavs Hospital Fund for Heart Research (O.J.K., U.W.), the European Union (FP6 Grant LSHM-CT-2005-018833; G.C.), the National Institutes of Health (HL078797-01A1; G.C.), the Italian Ministry of Health and Education (G.C.), and the British Heart Foundation (G.L.S.).

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